



COMSATS University Islamabad, Attock Campus

Department of Computer Science

Semester: Spring 2026

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Assignment 3

TASK 1 SS

```
LAB ASSIGNMENT3.py > ...
1  import hashlib, datetime
2
3  class Block:
4      def __init__(self, i, data, prev_hash):
5          self.i = i
6          self.t = datetime.datetime.now()
7          self.data = data
8          self.prev = prev_hash
9          self.hash = self.calc()
10
11     def calc(self):
12         s = f"{self.i}{self.t}{self.data}{self.prev}"
13         return hashlib.sha256(s.encode()).hexdigest()
14
15     class Blockchain:
16         def __init__(self):
17             self.chain = [Block(0, "Genesis", "0")]
18
19         def add(self, data):
20             prev = self.chain[-1]
21             self.chain.append(Block(prev.i+1, data, prev.hash))
22
23         def show(self):
24             for b in self.chain:
25                 print(b.i, b.data, b.hash[:10], "...")
26
27     # Test
28     bc = Blockchain()
29     for i in range(1,5):
30         bc.add(f"Block {i}")
31
32     bc.show()
33
34     # Tamper
35     print("\nTampered:\n")
36     bc.chain[2].data = "Hacked"
37     bc.chain[2].hash = bc.chain[2].calc()
38     bc.show()
```

OUTPUT:

```
PS C:\Users\zakar\OneDrive\Desktop\IS_LAB_ASSIGNMENT3_4> & C:\Users\zakar\AppData\Local\Programs\Python\Python314\python.exe "c:\Users\zakar\OneDrive\Desktop\IS_LAB_ASSIGNMENT3_4\LAB_ASSIGNMENT3.py"
0 Genesis: a668f730d ...
1 Block 1: 40000ac19b ...
2 Block 2: 9cd02058ef ...
3 Block 3: 4f0bfcdd3 ...
4 Block 4: 1b4d0fc094 ...

Tampered:
0 Genesis: a668f730d ...
1 Block 1: 40000ac19b ...
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PS C:\Users\zakar\OneDrive\Desktop\IS_LAB_ASSIGNMENT3_4>
```

TASK 2 SS

```
LAB ASSIGNMENTS_2.py ? ...
1 # Curve params
2 a, p = 2, 97
3 O = (None, None)
4
5 def inv(x): return pow(x, -1, p)
6
7 def add(P, Q):
8     if P == O: return Q
9     if Q == O: return P
10    x1,y1 = P; x2,y2 = Q
11
12    if P == Q:
13        m = (3*x1*x1 + a) * inv(2*y1) % p
14    else:
15        if x1 == x2: return O
16        m = (y2-y1) * inv(x2-x1) % p
17
18    x = (m*m - x1 - x2) % p
19    y = (m*(x1 - x) - y1) % p
20    return (x,y)
21
22 def mul(k, P):
23     R = O
24     while k:
25         if k & 1: R = add(R, P)
26         P = add(P, P)
27         k >>= 1
28     return R
29
30 # Keys
31 G = (3,5)
32 priv = 5
33 pub = mul(priv, G)
34
35 # Encrypt
36 msg = 10
37 k = 7
38 C1 = mul(k, G)
39 C2 = (msg + mul(k, pub)[0]) % p
40
41 # Decrypt
42 dec = (C2 - mul(priv, C1)[0]) % p
43
44 print("Public:", pub)
45 print("Encrypted:", C1, C2)
46 print("Decrypted:", dec)
```

OUTPUT

```
PS C:\Users\zakar\OneDrive\Desktop\IS_LAB_ASSIGNMENT3_4> & C:\Users\zakar\AppData\Local\Programs\Python\Python314\python.exe "c:\Users\zakar\OneDrive\Desktop\IS_LAB_ASSIGNMENT3_4\LAB_ASSIGNMENT3.py"
0 Genesis: a668f730d ...
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PS C:\Users\zakar\OneDrive\Desktop\IS_LAB_ASSIGNMENT3_4> & C:\Users\zakar\AppData\Local\Programs\Python\Python314\python.exe "c:\Users\zakar\OneDrive\Desktop\IS_LAB_ASSIGNMENT3_4\LAB_ASSIGNMENT3_2.py"
Traceback (most recent call last):
```